LICENSURE EXAMINATION FOR TEACHERS (LET)

WHAT TO EXPECT

FOCUS: Professional Education

Facilitating Learning

LET Competencies:

1. Analyze the cognitive, metacognitive, motivational and socio-cultural factors that affect learning
2. Organize the learning environment that promotes fairness regardless of culture, family background and gender, responsive to learner's needs and difficulties

PREPARED BY: ***Aggarao, Marivilla Lydia B.***

PART I: Content Update

Basic Concepts

Schema - Prior knowledge

Principle - Universal truths/facts

Theory

Public pronouncement of what a scientist or an independent/group of minds that have done something and is subject for further studies/research.

Concepts/propositions that help to describe and explain observations that one has made.

Learning

- involves acquisition of new elements of knowledge, skills, beliefs and specific behavior
- - may mean one or
- more of all these things:

the act of gaining knowledge **(to learn something),** the knowledge gained by virtue of that act **(that which is known)** the process of gaining knowledge **(learning how).** -***Banner and Cannon, 1997***

LEARNING - It is an ongoing process of continued ***adaptation*** to our environment, ***assimilation*** of new information and ***accommodation*** of new input to fit prior knowledge**.**

Adaptation - to become adjusted to new or different conditions

Assimilation - to make or become similar; to become absorbed, as knowledge

Accommodation - to settle; reconcile, adapt, adjust

Learning

- is characterized by:

- a change in behavior or the capacity to change one's behavior in the future
- a relatively permanent observable/demonstrable change in the behavior of a person as a result of interaction of the environment
- occurring through practice or experience
- it is not being the same as thinking as its focus is on manifest behavior rather than simply on

thoughts

Principles/Conditions of Learning

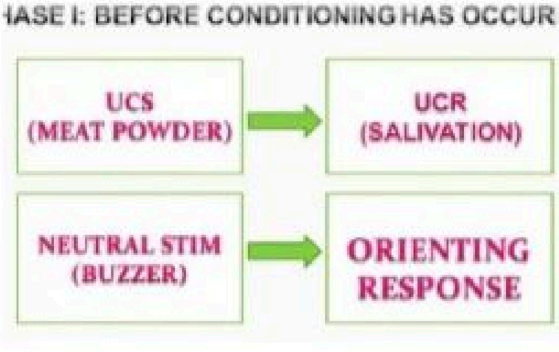
1. Learning is an active, continuous process: it involves more than acquiring information.

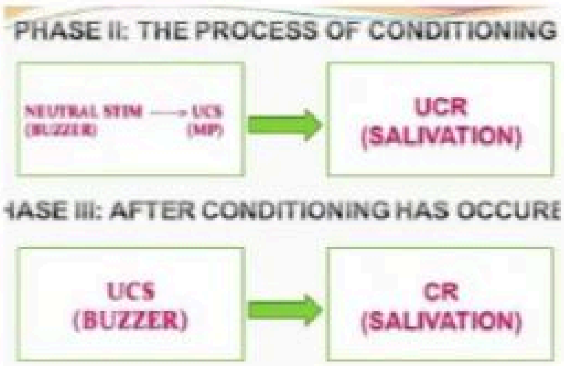
2. Styles and rates of learning vary: learners may be auditory, visual, or tactile/kinesthetic.
3. Readiness affects motivation and desire to learn.
4. Learning is very effective when there is immediate application of what is being taught.
5. Life experiences influence learning.
6. Learning is facilitated when learners have knowledge of their progress towards a goal.
7. Repetition (practice) helps perfect learning.
8. Principle of effect: learning is strengthened when accompanied by satisfying feeling.
9. Principle of primacy: what is taught must be taught right at the first time.
10. Principle of intensity: teaching requires provision of vivid, exciting learning of experiences.
11. Principle of recency: the things most recently learned are the best remembered.

Learning Theories
They are sets of ***conjectures*** and ***hypothesis*** that explain the process of learning or how learning takes place
Conjectures -to conclude or suppose from incomplete evidence; guess; an indecisive opinion
Hypothesis - a set of assumptions, provisionally accepted as a basis of reasoning or unsupported or ill supported theory

A. Behaviorism/Associative Learning Theory
It operates on a principle of "Stimulus-Response"
Prefers to concentrate on actual behavior
Ivan Petrovich Pavlov'sClassical Conditioning

- Classical means "in the established manner"
- Individual learns when a previously neutral stimulus is repeatedly paired with an unconditioned stimulus until a neutral stimulus evokes a conditioned response.





Feature of Classical Conditioning	Classroom Implications
1. **Stimulus – Generalization** – a process by which the conditioned response transfers to other stimuli that is similar to the original conditioned stimulus. Ex. stern teacher	• A child should be convinced that not all teachers in school are bad or anything that associates to school matters are bad.
2. Discrimination – a process by which one learns not to respond to similar stimuli in an identical manner because of previous experiences.	• Help the child to distinguish the difference between two or three identical stimuli or to discriminate their distinct differences.
3. Extinction – a process by which a conditioned response is lost. Ex. anxiousness	• Fear of anxiety towards a terror teacher gradually vanishes if in the succeeding days you experience pleasant treatment with the teacher.

Classroom Application
Relate learning activities with pleasant events.
Build positive associations between teaching and learning activities.

Edward Lee Thorndike's Connectionism
Connectionism means learning by selecting and connecting

Thorndike Theory of Learning	Classroom Implications
1. Multiple response – variation of responses that would lead to conclusion or arrival of an answer	✓ A child tries multiple responses to solve a certain problem.
2. Law of Set and Attitude – attitude means "disposition", "pre-judgment", and prior instruction/experience affects towards a given task.	✓ Giving of homework, advanced reading affects learning

3. Law of Readiness – interfering with oral directed behavior causes frustration, causing someone to do something they do not want to do is also frustrating. a. When someone is ready to perform some act, to do is satisfying. b. When someone is ready to perform some act, not to do is annoying. c. When someone is not ready to perform some act and is forced to do, it is annoying.	✓ Asking a child to write the alphabets when he/she did not learn the basic strokes of writing gets frustrated and annoyed.
4. Law of Exercise – the organism learns by doing and forgets by not doing. a. Law of use – connections between stimulus and response are strengthened as they are used. b. Law of disuse – connections between a stimulus and response are weakened when practice is discontinued.	✓ Practice makes perfect ✓ Provide varied enhancement activities/exercises, seatwork.
5. Law of Effect – reward increases the strength of a connection whereas punishment does nothing.	✓ Praise students' achievements; encourage those low performing students to do better.

Classroom Application
Do not force the child to go to school if he/she is not yet ready. Indications of readiness: sustained interest, improved performance (Ex. Writing, reading)
Practice what has been learned
Consider individual differences.

Burrhus Frederic Skinner's Operant Conditioning and Reinforcement
Operant Conditioning - using pleasant or unpleasant consequences to control the occurrence of behavior.
Reinforcers– any consequence that strengthen a behavior

- **Primary reinforcer**– related to basic needs. Ex. food
- **Secondary reinforcer** – value of something is acquired when associated with primary reinforcer. Ex. money to buy food
- **Positive reinforcer**– consequence given to strengthen a behavior
- **Negative reinforcer** – release from an unpleasant situation to strengthen behavior.

Reinforcement –it is a key element to explain why and how learning occurs.

- **Verbal** – praise, encouragement
- **Physical** – touch, pats, hugs
- **Non-verbal** – smiles, winks, warm looks
- **Activity** – games, enjoyments
- **Token** – points, stars

- ❖ **Consumable** – cookies

Punishment – any unpleasant consequence to weaken a behavior
Classroom Application
Teachers may use pleasant or unpleasant consequence to control the occurrence of behavior
Act on a situation right away. Be sure to make students understand why they are being reinforced or punished

B. Cognitive and Metacognition
Main focus is on memory (the storage and retrieval of information)
Prefer to concentrate on analyzing cognitive processes
Believe in the non-observable behavior

Basic Concepts:
1. Perception - a person's interpretation of stimuli.
2. Encoding – putting information in memory
3. Storage – changing the format of new information as it is being stored in memory
4. Rehearsal – mental repetition of information
5. Dual Coding – holds the complex networks or verbal representations and images to promote long term retention.
6. Retrieval – finding information previously stored in memory; recalling

Meaningful learning occurs when new experiences are related to what a learner already knows.

May occur through:

- reception
- rote learning
- discovery learning

David Ausubel's Meaningful Reception Theory

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Two Dimensions of Learning Processes:

The first dimension relates to the two ways by which knowledge to be learned is made available to the learner	**The second dimension relates to the two ways by which the learner incorporate new information into his existing cognitive structure**

1. Meaningful Reception Learning	3. Meaningful Discovery Learning
2. Rote Reception Learning	4. Rote Discovery Learning

Meaningful Reception Learning
material is presented to the learner in a well-organized/final form and relates it to his/her existing knowledge

Rote Reception Learning
material is presented to the learner in a well-organized/final form and is memorized

Meaningful Discovery Learning
learner arrives at the solution to a problem or other outcome independently and relates it to his/her existing knowledge.

Rote Discovery Learning
the solution is arrived at independently but is committed to memory

Classroom Application
Teachers to take note that before actual learning is expected, the teachers may use advance organizers

Jerome Bruner's Discovery Learning Theory or Inquiry Method/Theory of Instruction

Posits that learning is more meaningful to learners when they have the opportunity to discover on their own the relationships among the concepts or to actively search for a solution to a problem

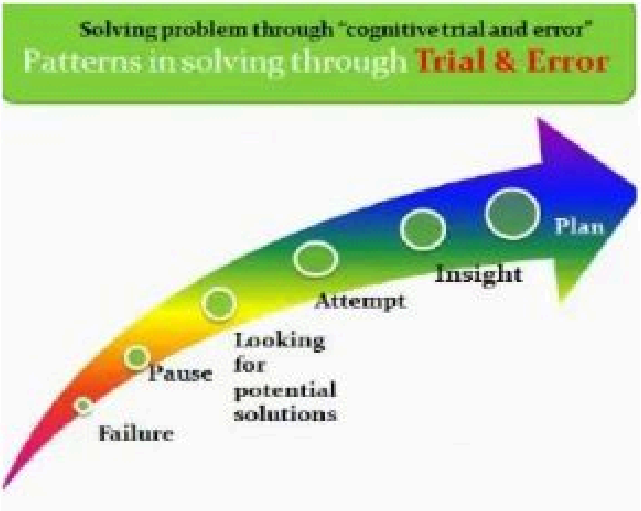
An approach to instruction through which students interact with their environment by exploring and manipulating objects, wrestling with questions and controversies or performing experiments. The idea is that students are more likely to remember concepts they discover on their own.
Calls his view of learning "instrumental conceptualism"

Scaffolding

Classroom Application
Teachers must strive to see a problem as the learner sees it and provide information that is consistent with learner's perspective.

Wolfgang Kohler's Insight Learning/Problem – Solving Theory
Insight – the capacity to discern the true nature of situation
- imaginative power to see into and understand immediately
- Gaining insight is a gradual process of exploring, analyzing, and structuring perception until a solution is arrived at.

The more intelligent a person and the more experiences he has, the more capable he will be for gaining insight.
Held that animals and human beings are capable of seeing relationships between objects and events and act accordingly to achieve their needs.
The power of looking into relationships involved in a problem and in coming up with a solution



Classroom Application
Allow students to go through trial and error method especially in doing laboratory experiments and in solving mathematical equations
Teachers should help students in gaining insights by giving/presenting activities/situations to do so, they will be able to solve their problems.

Jean Piaget's Cognitive Constructivism

- It emphasizes the active role of the learner in building understanding and making sense of information.
- It is about how the individual learner understands things, in terms of developmental stages and learning styles

Two major parts:

1. **Ages** – what children can and cannot understand at different ages
2. **Stages** – how children develop cognitive abilities through developmental stages

Developmental Stages - it is a distinct period in the life cycle characterized by a particular sets of abilities, motives, behavior and emotion that occur together and form a coherent pattern.
Classroom Application
Consider the developmental stages and learning styles of learners in presenting ideas
Teachers should provide necessary resources and rich environment filled with interesting things to explore, thus become active instructor of their own knowledge

Richard Atkinson's and Richard Shiffrin's Information Processing Theory
The individual learns when the human mind takes in information (**encoding**), performs operation in it, stores the information (**storage**), and retrieves it when needed (**retrieval**)
Memory – the ability to store information so that it can be used at a later time.

Stages of Human Memory
1. Sensory Memory – utilizes sense organs such as visual, auditory; lasts less than a second
Ex. color, shape, blowing of horn
2. Short Term Memory (STM) – selected by attention; lasts up to 13-30 seconds
Ex. telephone number
3. Long Term Memory (LTM) – lasting retention of information

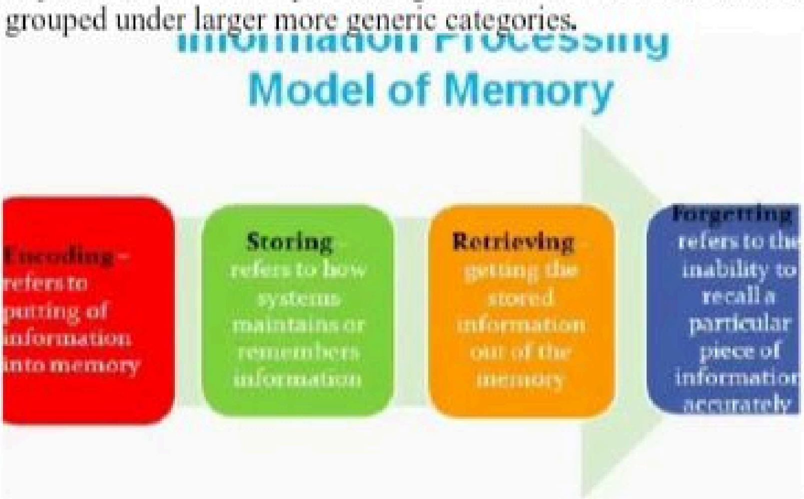
- Minutes to lifetime
- Information on The LTM, if not rehearsed, can be forgotten through trace decay

Three components:

Episodic Memory – associated with our recall of particular times and places and a storage place for many personal experiences.

Procedural Memory – refers to "knowing how" as opposed to "knowing that"

Semantic Memory – knowledge of general facts, principles and concepts that are not connected to particular times and places.Organized in networks of connected ideas or relationships referred grouped under larger more generic categories.



Forgetting

To be unable to recall (something previously known) to the mind

Causes of Forgetting

1. Retrieval Failure- forgetting is due to inability to recall the information.
2. Decay Theory – information stored in LTM gradually fades when it is not used.
3. Interference Theory – forgetting in LTM is due to the influence of other learning

Retention- the ability to recall or recognize what has been learned or experienced.

Interference – the act or an instance of hindering, obstructing or impeding.

Classroom Application

Hold learner's attention in all cognitive tasks.

Assist learner to assess materials considered most important to learn

Employ interesting rehearsal activities

Organize information to be learned

Robert Gagne's Cumulative Learning

Learning skills are hierarchically arranged

Progression from stimulus-response associations to concepts, principles and problem solving.

7 Levels of Learning

1. Signal Learning – involuntary responses are learned Ex. hot surface touched
2. Stimulus-response Learning – voluntary responses are learned. Ex. Getting ready to move at the sound of a fire alarm
3. Chaining/Motor – two or more separate motor/verbal responses maybe combined or chained to develop a more complex shell Ex. house + wife = housewife
4. Discrimination Learning – learner selects a response which applies to stimuli. Ex. sound of fire engine is different from other sounds/sirens
5. Concept Learning – involves classifying and organizing perceptions to gain meaningful concepts Ex. Concept of "triangle", discriminate triangle from other shapes and deduce commonality among different shapes
6. Principle Learning (Rule Learning) – involves combining and relating concepts to form rules Ex. Equilateral triangles are similar in shapes
7. Problem Solving – considered the most complex condition: involves applying rules to appropriate problem situations Ex. Solving mathematical problems using a given formula (find the area of a square A = 1xW)

Teaching for Transfer (Gagne)
Transfer – to convey or cause to pass from one place, person or things to another; direct (a person) elsewhere for help or information.

Transfer of Learning
Types:
1. **Lateral transfer** – occurs when the individual is able to perform a new task about the same level. (e.g. solving word problems given in text and later solving a similar problem on the board)
2. **Vertical transfer** – occurs when the individual is able to learn more advanced/complex skills (e.g. being able to add and multiply; being able to read and write)

Classroom Application
Observe strictly sequence in teaching in terms of level of learning skills and capabilities required.
Check students' capabilities in each level before moving to the next level.
Make sure that pre-requisite learning is required before proceeding to the target level.

Kurt Lewin's Field Theory

- Known for the terms: "life space" (reality, need, aspirations, desires, goals) and "field theory" (forces-social environment; function of both the person and environment
- "Learning is more effective when it is an active rather than a passive process"

Classroom Application
In a classroom for instance teachers must try to suit the goals of the activities of the lessons to the learner's needs along with his environment.

C. Socio-cultural
Concepts:
Learning involves participation in a community of practice
Society and culture affects learning
Social learners become involved in a community of practice, which embodies certain beliefs and behaviors to be acquired; social interaction.

Culture and Learner Diversity
Relationship of culture and learning style affect students' learning/achievement.
* student's color, "way of life" vs. cultural values, beliefs and norms of schools
Teacher's cultures
- teacher's own cultural orientations impede successful learners guided by another cultural orientation.

Albert Bandura's Social / Observational Learning Theory
Known for his "Bobo doll" experiment
People learn through observation, simulation, modeling which means watching (observing), another called a model and later imitating the model's behavior.
Concentrates on the power of example

Models are classified as:
Real life– exemplified by teachers, parents and significant others
Symbolic – presented through oral/written symbols, e.g. books
Representational– presented through audio-visual measures, e.g. films

Concepts in Social Learning Theory
1. Vicarious Learning – learning is acquired from observing the consequences of other's behavior.
2. Self – regulated Learning – occurs when individuals observe, assess and judge their own behavior against their own standards, and subsequently reward or punish themselves.
4 Phases of Observational Learning
1. Attention – mere exposure does not ensure acquisition of behavior. Observer must attend to recognize the distinctive features of the model's response.
2. Retention –reproduction of the desired behavior implies that student symbolically retains that observed behavior

3. Motor Reproduction Process– after observation, physical skills and coordination are needed for reproduction of the behavior learned.
4. Motivation al Process– although observer acquires and retains ability to perform the modeled behavior, there will be no overt performance unless conditions are favorable

Classroom application:
Model desirable behaviors, making sure that the students are paying attention while doing so
Make sure that the students are physically capable of doing the modeled behavior and that they know why they should demonstrate this behavior
Expose students to a variety of exemplary models

Situated Learning by Jean Lave and Wenger
Concepts
The nature of the situation impacts significantly on the process of learning.
Learning involves social relationships – situations of co-participation.
Learning is in the relationships between people.
Learning does not belong to individual persons, but to the various conversations they share.
Classroom Application
Engage students in group activities/participatory works
Allow students to do/participate in community – based activities
Relate teaching- learning to real life situations

UrieBrofenbrenner's Ecological Systems Theory/Environmental Contexts
Learning is greatly affected by the kind of environment we are in.
Learners are understood within the context of their environment.
These environmental contexts are interrelated.

Environmental Contexts: Major Levels
1. Microsystem – innermost level
- contains the structure that has direct contact with child

2.Mesosystem– connection between the structures of the child's microsystem
3.Exosystem – 3rd level
- social system which the child does not function directly

4.Macrosystem– outermost level
- values, customs, laws, beliefs and resources of a culture/society

5 . Chronosystem–

If the relationships in the immediate microsystem break down, the child will not have the tools to explore other parts of his environment resulting to behavioral deficiencies. Learning tends to regress / slow down when the environment of the child is in turmoil

Classroom Application

School and teachers should work to support primary needs of the learner to create an environment that welcomes and nurtures school – home relationship through: parent-teacher conferencing, home visitation, telephone brigade, family day

Lev Vygotsky's Social Constructivism
It emphasizes how meaning and understanding grow out of social encounters.

Zone of Proximal Development (ZPD) - gap between actual and potential development
*Actual development – what children can do on their own
* Potential development – what children can do with help

Scaffolding –

- competent assistance or support through mediation of the environment (significant others) in which cognitive, socio-emotional and behavioral development can occur.

Classroom Application
Engage students in group activities and let them share their schema on a particular subject within the groups (small groups) and synthesize it in the big group

Howard Gardner's Multiple Intelligences
Intelligence – refers to general mental ability of a person
- capacity to resolve problems or to fashion products that are valued in a more cultural setting

Achievements – refers to the previous learning of a person in a certain subject area.

Multiple Intelligence – capacity of a person to possess and adapt two or more intelligences.

Intelligence	**Competence**	**Examples**
1. Linguistic – sensitivity to spoken and written language	- Ability to learn language - Capacity to use language to accomplish certain goals	▪ Writers, poets, lawyers, speakers
2.Logical/mathematical – analyzes problems logically, carry out mathematical operations, and investigate issues scientifically.	- Ability to detect patters, reason deductively and think logically.	▪ Scientists, mathematicians
3. Musical – skill in the performance, composition and appreciation of musical patterns.	- Capacity to recognize and compose musical pitches, tones and rhythms.	▪ Musicians, composers
4. Bodily kinesthetic – using one's whole body or body parts to solve and convey ideas.	- Ability to use mental abilities to coordinate bodily movements.	▪ Athletes, dancers

5. Spatial – recognize and use patterns of wide space and more confined areas.	Capacity to understand, appreciate and maximize the use of spaces	• Engineers
6. Interpersonal – working effectively with others.	- Capacity to understand the intentions, motivations and desires of other people.	▪ Educators, sales people, religious counselors, politicians
7. Intrapersonal – working effectively with oneself	- Capacity to understand oneself, appreciate one's feelings, fears and motivations	
8. Naturalist – appreciation of the environment/nature.	- Ability to recognize, categorize and grow upon certain features of the environment	▪ Nature lover, environmentalist

Classroom Application
Make use of various activities which will address the different intelligences of your students in the class (e.g. art activities to accommodate art inclined students, song writing for musically inclined, etc)

Robert Sternberg Triarchic Intelligence (1988), focuses on three main components of intelligence:

Practical intelligence--the ability to do well in informal and formal educational settings; adapting to and shaping one's environment; street smarts.
Experiential intelligence--the ability to deal with novel situations; the ability to effectively automate ways of dealing with novel situations so they are easily handled in the future; the ability to think in novel ways.
Componential intelligence--the ability to process information effectively.Includes metacognitive, executive, performance, and knowledge-acquisition components that help to steer cognitive processes.
Classroom Application
Engage students in practical, experiential and classroom-based activities.

Daniel Goleman's Emotional Intelligence
Highlights the role of emotion in the success or happiness of an individual which eventually affects behavior or learning.

Classroom Application
Surface the emotions manifested by students in a certain situation. Don't suppress it.

William Glasser's Control Theory
Behavior is inspired by what satisfies a person's want at any given time.

Classroom Application
Make schoolwork relevant to student's basic human needs.

David Kolb's Learning Styles
Learning Styles – are tools utilized by learners to cope and adjust to the learning environment

Four learning styles

Learning Styles	**Educational Implications**
1. Convergers– rely on abstract conceptualizing and experimenting - they like to find specific, concrete answers and move quickly to solution - unemotional, since they prefer to deal with things rather than with people. Ex. physical sciences and engineering	▪ Teacher should provide learning tasks that have specific answers like numbers and figures/units.
2. Assimilators – rely most on abstract conceptualizing and reflective observation - interested in theoretical concerns than in applications. Ex. research and planning	▪ Teacher should provide learning tasks that call for integration of materials/situational activities
3. Divergers – rely on concrete experience and active participation - generate ideas and enjoy working with people. Ex. counseling and consulting	▪ Teacher should provide group activities since learners enjoy working in groups.
4. Accommodators – rely on concrete experience and active experimentation - risk – taking, action oriented, adoptable in new situations. Ex. marketing, business, sales	▪ Teacher should provide learning tasks that call for hands-on approach.

Types of Learners

Types of Learners/Perceptual Channel	**Educational Implications/Learning Preferences**
1. Auditory learners – prefer to learn by listening/auditory perceptual channel.	▪ Lecturing is the teaching approach that works best for them. ▪ Songs/poems are useful and effective learning tools.

2. Visual learners – prefer print materials/visual perceptual channel	▪ Reading/responding to visual cues, such as the chalkboard or transparencies ▪ Textbooks and pictures are useful and effective learning tools.
3. Tactile learners – like to manipulate objects/tactile perceptual channel	▪ Hands-on or laboratory methods of learning are most appropriate for learners. ▪ Tracing diagrams or using texture examples.
4. Kinesthetic or whole body learners – like to learn through experiential activities/kinesthetic perceptual channel.	▪ Simulations, exploratory activities and problem-solving approach of teaching. ▪ Pacing or dancing while learning new material.

Part II
MOTIVATION

What is meant by Motivation?
An internal state or condition (sometimes described as a need, desire or want) that serves to activate or energize behavior and give it direction.
Although motivation cannot be seen directly, it can be inferred from behavior we ordinarily refer to as ability. Ability refers to what a person wants to do.
In order to do this effectively, it is necessary to understand that motivation comes in two forms.

Two Kinds of Motivation:

Extrinsic Motivation
– When students work hard to win their parents' favour, gain teachers' praise or earn high grades; their reasons for work and study lie primarily outside themselves.
- Is fuelled by the anticipation and expectation of some kind of payoff from an external source

Intrinsic Motivation
– when students study because they enjoy the subject and desire to learn it, irrespective of the praise won or grades earned; the reasons for learning reside primarily inside themselves
- Fuelled by one's own goal or ambitions

Principles of Motivation
The environment can be used to focus the student's attention on what needs to be learned.
Incentives motivate learning

Internal motivation is longer lasting and more self – directive than is external motivation, which must be repeatedly reinforced by praise or concrete rewards.
Learning is most effective when an individual is ready to learn, that is when one want to know something.
Motivation is enhanced by the way in which the instructional material is organized.

Theories of Motivation

1. **Drive Theory (Clark Hull)**
    - Drive is a condition of arousal on tension that motivates behavior
    - Drives most typically have been considered to involve physiological survival needs; hunger, thirst, sleep, pain, sex.
    - A drive results from the activation of a need
    - Need – a physiological deficiency that creates condition of disequilibrium in the body

2.Self – Efficacy (Albert Bandura)
Self – efficacy – it is the belief that one has capabilities to execute the courses of actions required to manage prospective situations. Unlike efficacy, which is the power to produce an effect (in essence competence) self – efficacy is the belief (whether or not accurate) that one has the power to produce that effect.

Self – efficacy relates to a person's perception of his/her ability to reach a goal while, self – esteem relates to a person's sense of self – worth.

3. Self – Determination (E. Deci)
Self – determination – comes from the sense of autonomy that a person has when it comes to things that he does and the choices he makes.

4. Theory of Achievement Motivation (Atkinson)
- Motivation to perform is affected by two variables
    - **Expectancy –** people must believe than they can accomplish a task, that is, they should have expectancy about what they want to achieve.
    - **Value –** they should place an importance or value in what they are doing.

5. Attribution Theory (B. Weiner)
People's various explanations for successes and failures – their beliefs about what causes attributions.
Dimensions underlying people's attribution. People can explain events in many different ways. For example, a tennis player may attribute his/her wins and successes in matches to things like – luck, health, effort, mood, strengths and weaknesses of his/her opponents, climate, his/her fans etc.

TECHNIQUES IN MOTIVATING LEARNERS

Challenge them - offer student's opportunities to undertake real challenges. Encourage them to take intellectual risks.
Build on strengths first - Opportunity to use their talents to achieve success.
Offer choices - offering choices develop ownership. When child makes decisions he/she is more likely to accept ownership and control of the results.
Provide a secure environment which permits children to fail without penalty. Learning how to deal with failure is critical for developing motivation and successful learning.

III. ACHIEVING LEARNING OUTCOMES

A. Definitions
- Learning outcomes specify what a learner is expected to know, understand or to be able to do as a result of a learning process.
- Measuring learning outcomes provides information on what particular knowledge (cognitive); skill or behavior (psychomotor and affective). Students have gained after instruction is completed.

B. Importance
- Communicate expectations to learners
- Review curriculum and content
- Design appropriate assessment
- Evaluate the effectiveness of learning

C. Three learning domains (KSA)
C.1. Cognitive Learning Domain
– development of knowledge and intellectual skills
- mental skills (knowledge)

Basic Concepts: Cognitive Learning
1.Fact – something that is true, something that actually exists
2. Concept – basically the main idea
3. Generalization –the formation of a general notion by putting together general concepts
4. Thinking – rational; reasoning

Types of Thinking
1 Problem Solving – process involved in the solution of a problem.

2. Critical Thinking
a. Careful and deliberate determination of whether to accept, reject, suspend judgement on a claim
b. Reasonable reflective thinking that is focused in deciding whether to believe or do

c. Comprises the mental processes, strategies and representations people use to solve problems, make decisions, and learn new concepts

3. Creative Thinking

- Involves the ability to produce new forms in an art or mechanics or to solve problems by novel methods
- Creativity consist in coming up with a new and relevant ideas
- Creativity has two kinds

a. Cognitive – involved in problem solving
b. Aesthetic – relating to artistic creation

4. Metacognition

-meta– after; beyond; higher
-**cognition**– way of thinking; perceiving; knowing

- Refers to the idea of "knowing about knowing", involves the study of how we think about our own thinking in order to develop strategies for learning.
- Is the capacity to monitor and regulate one's own thinking or mental capacity.
- From of thinking in which an individual develops an awareness of his characteristics, attitudes, beliefs, and actions.

Principles in Achieving Cognitive Learning and Their Classroom Implications

- **Content:** Teach tacit heuristic knowledge as well as textbook knowledge.
- **Situated Learning:** Teach knowledge and skills that reflect the way the knowledge will be useful in real life.
- **Modeling and Explaining:** Show how a process unfolds and tell reasons why it happens that way.
- **Coaching and Feedback:** Pay personalized attention to performance, coupled with appropriate hints, helps, and encouraging feedback.
- **Articulation and Reflection:** make students think about and give reasons for their actions/own performance.
- **Exploration:** Encourage students to try out different strategies and observe their effects.
- **Sequence:** Proceed in an order from simple to complex, with increasingly diversity.

C.2. Affective Learning Domain

(Krathwol)

- deals with attitudes, motivation, willingness to participate
- valuing what is being learned
- incorporating the values of a discipline as a way of life
- growth in feeling or emotional areas (attitude)

Basic Concepts: Affective Learning

- **Beliefs –** an accepting of something or someone as true or reliable without asking for proof.

- **Attitudes –** a particular feeling or way of thinking about something.
- **Values –** important and enduring beliefs or ideals shared by the members of a culture about what is good or desirable and what is not.

Principles in Achieving the Development of Attitudes and Values and Their Classroom Implications

- Every interaction with children provides an opportunity to teach values.
- Children learn about our values through daily interaction with us.
- Children learn through our example
- Children learn values through the way we do things as a family.
- Children learn values and beliefs through their exposure to the larger world.
- Children learn values through our explanations of the world.

C.3 Psychomotor Learning Domain
(Anita J. Harrow)
Includes physical movement that involves coordination of the mind and body
Manual of physical skills

Basic Concepts: Psychomotor Learning
Capacity – the facility or power to produce, perform or deploy.
Ability – competence in an activity or occupation because of ones' skill, training, or other qualification.
Skill – learned capacity to carry out predetermined results often with the minimum outlay of time, energy, or both.

PRINCIPLES INVOLVED IN ACHIEVING PSYCHOMOTOR LEARNING AND THEIR CLASSROOM IMPLICATIONS
1. The psychomotor domain is best assessed in a face to face situation.
2. It focuses on performing sequences of motor activities to a specified level of motor operations for a child of given age.
3. Learning materials and activities should involve the appropriate level of motor capabilities.
4. Use teaching methods that actively involve students and present challenges.
5. Psychomotor learning is facilitated by providing activities or situations that engage learners to perform.

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